

Amedeo Carraro

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Computer Engineer specialized in Artificial Intelligence and Machine Learning, experienced in both deep learning architectures and generative AI systems. Skilled in bringing AI solutions from initial concept through to production deployment, working with cross-functional teams to deliver scalable applications with measurable results. Want to learn more about me or my projects? Visit my website and ask my AI personal assistant!

Professional Experience

AI and Machine Learning Engineer, Terya Spa, Bassano del Grappa, Italy Nov 2025 – Ongoing

- Designed and deployed production-ready conversational AI solutions including RAG-based chatbots and autonomous agents, implementing hybrid semantic and full-text search pipelines with Elasticsearch, custom document processing workflows, and LLM-powered response generation, fully containerized with Docker
- Developed a real-time voice assistant integrated with enterprise telephony infrastructure for phone support, leveraging OpenAI Realtime API for bidirectional voice conversations, audio preprocessing, automated ticket creation via REST API integration with service desk platform, and call routing with escalation to operators

Education

MSc in Computer Engineering: Artificial Intelligence and Robotics, University of Padua Sep 2022 – Mar 2025

Erasmus+ University of Cyprus (2024): Natural Language Processing (NLP), AI in Medicine, Entrepreneur AI

Relevant Modules: Machine and Deep Learning, Artificial Intelligence, Computer Vision, Intelligent Robotics

Research Training Intern: Neural Networks for Variable Star Classification Sep 2024 – Mar 2025

- Implemented two hybrid deep learning architectures (2D-CNN and CNN-LSTM) for classifying variable stars from time-series photometric data, synthesizing insights from multiple peer-reviewed publications to optimize model performance for large-scale astronomical surveys
- Achieved strong classification performance with F1-scores consistently exceeding 95% across stellar categories and peaks at 99% on the complete Zwicky Transient Facility dataset

Full Stack Developer Intern, ITReview, Padua, Italy Nov 2023 – Dec 2023

- Developed responsive web applications with React frontend and Node.js/Express backend, integrating RESTful APIs and managing data persistence across PostgreSQL (raw SQL via pg client) and MongoDB (Mongoose)
- Participated in code reviews and performance optimization initiatives, implementing indexing strategies and query refactoring that reduced database response times by approximately 15%

BSc in Computer Engineering, University of Padua Sep 2016 – Mar 2022

Relevant Modules: Software Engineering, Databases, Operating Systems, Data Structures & Algorithms

Deep Learning & DevOps Intern, AppLibra s.r.l, Thiene, Italy Oct 2021 – Dec 2021

- Developed and deployed a real-time computer vision system using YOLOv5 (PyTorch) and OpenCV to detect bottle cap defects on production conveyor belts, handling the full pipeline from dataset annotation with LabelImg to training and real-time inference on Docker-based Linux environment
- Configured a BaGet server in Docker as private NuGet repository for .NET teams, standardizing library distribution and version control across projects and eliminating manual dependency management prone to versioning conflicts
- Contributed to ERP migration from Microsoft Dynamics Navision to Dynamics 365 Business Central, developing custom pages and forms using C/AL (C/SIDE) and AL (VSCode), and building XMLports for structured data import/export to ensure accurate data migration across client organizations

Technical Skills and Languages

- Programming Language: Python, Java, C, C++, JavaScript, C#, SQL, HTML, PHP, MATLAB
- Frameworks: TensorFlow, Numpy, Pandas, Scipy, Matplotlib, Keras, PyTorch, scikit-learn, GitHub
- Languages: Italian (mother tongue), English (C1) and French (A2)
- Certification: OCI AI Foundations Associate, OCI Generative AI Professional, OCI AI Vector Search Professional

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